Type 2 Diabetes

Patient Education Guide

What Is Type 2 Diabetes?

Type 2 diabetes is a chronic condition that affects how the body processes blood glucose (blood sugar). In type 2 diabetes, the body either becomes resistant to the effects of insulin — a hormone produced by the pancreas that moves glucose from the bloodstream into cells for energy — or does not produce enough insulin to keep blood glucose in a normal range. Over time, this insulin resistance and relative insulin shortage cause glucose to build up in the blood instead of being used for energy.

Type 2 diabetes accounts for roughly 90–95% of all diagnosed diabetes cases, making it by far the most common form of the disease. It typically develops gradually over months or years and was once seen mainly in adults over 45, though it is increasingly diagnosed in younger adults and adolescents, largely alongside rising rates of obesity and sedentary lifestyles. Unlike type 1 diabetes — an autoimmune condition usually diagnosed in childhood — type 2 diabetes is strongly influenced by lifestyle and metabolic factors, which means it can often be prevented, delayed, or well managed through diet, physical activity, and, when needed, medication.

Risk Factors

Several factors increase the likelihood of developing type 2 diabetes. Many can be modified through lifestyle change, while others — such as genetics — cannot:

- **Family history and genetics** — having a parent or sibling with type 2 diabetes significantly raises personal risk.
- **Excess body weight** — obesity, particularly abdominal fat, is one of the strongest risk factors, as it increases insulin resistance.
- **Sedentary lifestyle** — low physical activity reduces the body's ability to use insulin effectively.
- **Age** — risk rises after 45, although diagnoses in younger people are increasing.
- **Prediabetes** — blood sugar higher than normal but not yet in the diabetes range.
- **Gestational diabetes history** — women who had diabetes during pregnancy face a higher lifetime risk.
- **Ethnicity** — South Asian, African, Hispanic, and Native American populations have a higher predisposition.
- **Related conditions** — high blood pressure and abnormal cholesterol frequently occur alongside type 2 diabetes and compound overall risk.

Symptoms

Type 2 diabetes often develops slowly, and many people have no noticeable symptoms in the early stages — which is why routine screening matters, especially for those with risk factors. When symptoms do appear, they commonly include:

- Polyuria — frequent urination, including at night
- Polydipsia — excessive thirst
- Persistent fatigue
- Blurred vision
- Slow-healing cuts, wounds, or frequent infections

- Increased hunger, sometimes with unintended weight loss
- Tingling, numbness, or pain in the hands or feet

Because symptoms can be subtle or absent, many people live with undiagnosed diabetes for years. Anyone with several risk factors should discuss screening with a healthcare provider even without symptoms.

Diagnosis Criteria

Diabetes is diagnosed using standardized blood tests. Any one of the following, typically confirmed on a separate day, is sufficient for diagnosis:

- **Fasting Blood Sugar (FBS):** 126 mg/dL or higher, after at least 8 hours without food
- **HbA1c:** 6.5% or higher — reflects average blood sugar over roughly the past 2–3 months
- **Oral Glucose Tolerance Test (OGTT):** 200 mg/dL or higher, 2 hours after a standardized glucose drink
- **Random blood glucose:** 200 mg/dL or higher, together with classic symptoms of high blood sugar

Prediabetes is diagnosed when values fall below the diabetes threshold but above normal: FBS of 100–125 mg/dL, or HbA1c of 5.7–6.4%. Prediabetes is a warning sign — and an opportunity to prevent or delay progression to diabetes through lifestyle change.

Management

Type 2 diabetes is managed through a combination of lifestyle change, monitoring, and — for many people — medication. The goal is to keep blood sugar within a healthy target range and reduce the risk of long-term complications.

Diet

A balanced eating pattern emphasizing whole grains, vegetables, lean protein, and healthy fats — while limiting refined carbohydrates and added sugars — helps stabilize blood glucose. Portion control and consistent meal timing also support better control.

Exercise

Regular physical activity improves the body's sensitivity to insulin. Most guidelines recommend at least 150 minutes per week of moderate aerobic activity, such as brisk walking, combined with resistance or strength training on 2 or more days per week.

Medications

Metformin is typically the first medication prescribed, as it lowers glucose production by the liver and improves insulin sensitivity with a well-established safety profile. If blood sugar remains above target, additional medications may be added, including sulfonylureas, SGLT2 inhibitors, or GLP-1 receptor agonists. Some people eventually require insulin therapy, particularly if the pancreas's own insulin production declines over time. Medication choice should always be guided by a physician.

Monitoring

Regular self-monitoring of blood glucose, or use of a continuous glucose monitor, helps track day-to-day control and catch patterns early. HbA1c testing every 3 to 6 months gives a longer-term picture of control, and routine follow-up visits allow screening for early signs of complications.

Complications

Poorly controlled blood sugar over years can damage blood vessels and nerves throughout the body. Consistent management significantly reduces — though does not eliminate — the risk of the following:

- **Neuropathy** — nerve damage, most often in the hands and feet, causing pain, tingling, or numbness

- **Retinopathy** — damage to blood vessels in the eye, which can lead to vision loss if untreated
- **Nephropathy** — progressive kidney damage that can eventually lead to kidney failure
- **Cardiovascular disease** — elevated risk of heart attack, stroke, and other circulatory problems
- **Poor wound healing** and increased infection risk, particularly in the feet, which in severe cases can lead to ulcers or amputation

Regular screening — including eye exams, kidney function tests, and foot checks — allows many of these complications to be caught and managed early.

When to See a Doctor

- Persistent symptoms such as excessive thirst, frequent urination, or unexplained fatigue
- A family history of diabetes combined with other risk factors, even without symptoms, to discuss screening
- An existing diagnosis of prediabetes or diabetes, for scheduled follow-up and monitoring
- New vision changes, numbness or tingling in the extremities, or slow-healing wounds
- Symptoms of very high blood sugar (confusion, extreme thirst, rapid breathing) or very low blood sugar in those on medication (shakiness, sweating, confusion) — these need prompt medical attention

This guide is intended for general educational purposes only and does not replace individualized medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider regarding any medical condition.